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SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I Veeraja Jeevan (192526013 3) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Jeevan

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%	1	1	2	2
Method Selection	20%	1	2	2	1
Solution Process	25%	3	3	2	3
Accuracy of Calculation	20%	2	1	1	1
Interpretation of Results	15%	2	1	1	2

Total Marks 35 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculation
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretat

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ANALYTICAL PROBLEM SOLVING

Detailed step-by-step solutions for Questions 1–5

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Signature: __v jeevan _____date. 29/10/26_____

Objective of the assignment

The objective of this assignment is to apply IPv4 subnetting and IPv6 addressing concepts to practical network-design problems. The solutions identify network addresses, broadcast addresses, valid host ranges, subnet capacities, address allocation boundaries, and the correct representation of IPv6 addresses. Each problem is solved in a systematic way by writing the given information, selecting the appropriate formula, showing calculations, checking the result, and explaining its practical meaning.

Important formulas used

Host bits (IPv4) = 32 − prefix length. Total addresses = $2^{\text{host bits}}$. Usable hosts = $2^{\text{host bits}} - 2$.

Block size = 256 − the relevant subnet-mask octet. The first address of a subnet is its network address; the final address is its broadcast address. Ordinary IPv4 hosts use addresses between these two addresses.

For IPv6, the address length is 128 bits and it is written as eight 16-bit hexadecimal groups called hextets. A prefix length identifies the number of leftmost bits belonging to the network prefix.

QUESTION 1 — SUBNET THREE UNIVERSITY LABS

Problem statement and given information

A university has the IPv4 address block **192.168.20.0/24**. It must be divided among three computer laboratories. Lab A requires a /26 subnet, Lab B requires a /27 subnet, and Lab C requires a /28 subnet. The allocation must be valid, must not overlap, and must leave the remaining addresses available for future use.

Step 1 — Understand the starting block

The /24 prefix means that the first 24 bits identify the network and the remaining 8 bits are available for subnetting and host addresses. The subnet mask for /24 is 255.255.255.0. The final octet can therefore contain 256 possible values, from 0 to 255. The full block contains $2^8 = 256$ addresses.

Step 2 — Calculate the requirements for Lab A

Lab A uses /26. Host bits = 32 − 26 = 6. Total addresses = $2^6 = 64$. Usable hosts = 64 − 2 = 62. The corresponding mask is 255.255.255.192 because the last mask octet is 192. The block size is 256 − 192 = 64, so the network address must start at a multiple of 64 in the final octet.

To avoid wasting space and to keep the allocation easy to verify, allocate the largest required subnet first. Lab A can begin at 192.168.20.0 and occupy final-octet values 0 through 63. Its network address is 192.168.20.0, its broadcast address is 192.168.20.63, and its ordinary host range is 192.168.20.1 through 192.168.20.62.

QUESTION 1 — CONTINUED: SUBNET ALLOCATION AND VERIFICATION

Step 3 — Calculate the requirements for Lab B

Lab B uses /27, so the host-bit count is 32 − 27 = 5. The total number of addresses is 2^5 = 32, and the usable host count is 32 − 2 = 30. The subnet mask is 255.255.255.224. Its block size is 256 − 224 = 32. After Lab A ends at .63, the next available valid /27 boundary is .64.

Therefore, Lab B is assigned 192.168.20.64/27. Its address range is 192.168.20.64 to 192.168.20.95. The first address, .64, identifies the network; the final address, .95, is the broadcast address. The usable host range is 192.168.20.65 through 192.168.20.94.

Step 4 — Calculate the requirements for Lab C

Lab C uses /28. Host bits = 32 − 28 = 4. Total addresses = 2^4 = 16, and usable hosts = 16 − 2 = 14. The subnet mask is 255.255.255.240. Its block size is 256 − 240 = 16. The next available boundary after Lab B is .96, which is divisible by 16 and is therefore a valid /28 network boundary.

Lab C is assigned 192.168.20.96/28. It occupies .96 through .111. The network address is 192.168.20.96, the broadcast address is 192.168.20.111, and valid hosts range from 192.168.20.97 to 192.168.20.110.

Lab	Prefix and mask	Network address	Usable host range	Broadcast	Usable hosts
A	/26 255.255.255.192	192.168.20.0	192.168.20.1–.62	192.168.20.63	62
B	/27 255.255.255.224	192.168.20.64	192.168.20.65–.94	192.168.20.95	30
C	/28 255.255.255.240	192.168.20.96	192.168.20.97–.110	192.168.20.111	14

Step 5 — Verify that the subnets do not overlap

Lab A ends at .63 and Lab B starts at .64, so there is no overlap between A and B. Lab B ends at .95 and Lab C starts at .96, so there is no overlap between B and C. The boundaries are valid for their respective block sizes: .0 is divisible by 64, .64 is divisible by 32, and .96 is divisible by 16.

Step 6 — Calculate remaining addresses

The three laboratories use 64 + 32 + 16 = 112 addresses. The original /24 block contains 256 addresses, so 256 − 112 = 144 addresses remain unallocated. The remaining continuous range is 192.168.20.112 through 192.168.20.255. This space can support future labs, printers, wireless access points, or other network segments, provided future subnets are planned on suitable boundaries.

Conclusion

The allocation is valid and non-overlapping. Lab A supports 62 hosts, Lab B supports 30 hosts, and Lab C supports 14 hosts. Using the largest subnet first makes the allocation simpler and reduces the chance of creating unusable gaps. The unused range is preserved for future expansion.

QUESTION 2 — FIND THE SUBNET FOR TWO SYSTEMS

Problem statement and given information

System 1 has IP address **192.168.1.130** and System 2 has IP address **192.168.1.190**. Both systems use subnet mask **255.255.255.192**. The task is to determine the prefix, subnet boundaries, network address, broadcast address, usable host range, and whether both systems belong to the same subnet.

Step 1 — Convert the subnet mask to prefix notation

The first three octets of the mask are 255. Each 255 contains eight binary 1s, giving 24 network bits. The last octet, 192, is 11000000 in binary and contributes two additional network bits. Therefore the mask contains 26 network bits and is written as /26.

The complete mask in binary is 11111111.11111111.11111111.11000000. The 26 ones identify the network portion, and the remaining 6 zeros identify the host portion. This distinction is important because devices can communicate directly at Layer 2 only when their addresses are in the same local subnet, unless a router provides communication between different subnets.

Step 2 — Calculate the block size

The last mask octet is 192. Block size = 256 − 192 = 64. Thus the last-octet subnet boundaries are 0, 64, 128, and 192. The corresponding ranges are 0–63, 64–127, 128–191, and 192–255. Each range contains 64 addresses.

Subnet boundary	Address range	Network address	Broadcast address
1	192.168.1.0–.63	192.168.1.0	192.168.1.63
2	192.168.1.64–.127	192.168.1.64	192.168.1.127
3	192.168.1.128–.191	192.168.1.128	192.168.1.191
4	192.168.1.192–.255	192.168.1.192	192.168.1.255

Step 3 — Locate System 1

System 1 ends in .130. The value 130 lies between 128 and 191, so it belongs to the third subnet. Its network address is 192.168.1.128 and its broadcast address is 192.168.1.191. The usable host range is 192.168.1.129 through 192.168.1.190. Since .130 is neither the network address nor the broadcast address, it is a valid host address.

Step 4 — Locate System 2

System 2 ends in .190. The value 190 also lies between 128 and 191. It therefore belongs to the same subnet, 192.168.1.128/26. The address .190 is the last usable host address because the next address, .191, is the broadcast address. A device can use .190 as its own address, provided no other device has already been assigned that address.

QUESTION 2 — CONTINUED: HOST CAPACITY AND FINAL CHECK

Step 5 — Calculate the number of usable hosts

The prefix is /26, so the number of host bits is 32 − 26 = 6. The total number of addresses in this subnet is $2^6 = 64$. In a conventional IPv4 subnet, the first address is reserved as the network address and the last address is reserved as the broadcast address. Therefore, usable hosts = 64 − 2 = 62.

The result of 62 is the total usable host capacity of the subnet, not 62 extra addresses for each system. Both systems share the same pool of 62 usable addresses. Network administrators must ensure that each assigned address is unique and that the subnet mask is configured consistently on all devices.

Step 6 — Compare the two systems

Property	System 1	System 2
IP address	192.168.1.130	192.168.1.190
Subnet mask	255.255.255.192	255.255.255.192
Prefix	/26	/26
Network address	192.168.1.128	192.168.1.128
Usable host range	192.168.1.129–.190	192.168.1.129–.190
Broadcast address	192.168.1.191	192.168.1.191
Valid host?	Yes	Yes; last usable host
Usable capacity	62 in the subnet	62 in the subnet

Step 7 — Explain the practical meaning

Because both addresses are in the same /26 subnet, the systems can normally exchange local traffic without sending that traffic to a default gateway. The devices still require suitable network settings, such as a correct subnet mask, a functioning link, and no duplicate IP addresses. If the two systems were in different subnets, traffic would generally need to be forwarded by a router.

A common mistake is to assume that every address inside a numerical range can be assigned to a computer. In fact, the first and last addresses have special purposes in conventional IPv4 subnetting. For this example, .128 is the network identifier and .191 is the broadcast address. Neither should be assigned to an ordinary host.

Final answer

Both 192.168.1.130 and 192.168.1.190 belong to subnet **192.168.1.128/26**. The network address is 192.168.1.128, the broadcast address is 192.168.1.191, the valid host range is 192.168.1.129–192.168.1.190, and the subnet supports 62 usable hosts. Both supplied IP addresses are valid host addresses.

QUESTION 3 — ALLOCATE SUBNETS TO FOUR DEPARTMENTS

Problem statement and given information

An organization has the address block **172.16.0.0/16**. It needs separate networks for HR, Sales, IT, and Admin. The required prefixes are HR /25, Sales /26, IT /27, and Admin /28. The goal is to allocate each subnet correctly, show network and broadcast addresses, determine usable host ranges, avoid overlap, and calculate the remaining unused space.

Step 1 — Understand the address block

The /16 prefix means that 16 bits identify the network and 16 bits remain within the organization's address block. The subnet mask is 255.255.0.0. The total number of addresses is 2^{16} = 65,536, ranging from 172.16.0.0 through 172.16.255.255. Since the department requirements differ, variable-length subnet masking (VLSM) allows each department to receive a subnet of a suitable size.

Step 2 — Calculate HR subnet capacity

HR uses /25. Host bits = 32 − 25 = 7. Total addresses = 2^7 = 128. Usable hosts = 128 − 2 = 126. The mask is 255.255.255.128 and the block size in the final octet is 256 − 128 = 128. Starting from 172.16.0.0, HR occupies 172.16.0.0 through 172.16.0.127.

The network address is 172.16.0.0. The broadcast address is 172.16.0.127. The valid host range is 172.16.0.1 through 172.16.0.126. This gives HR 126 usable host addresses, which may be assigned to workstations, printers, and other devices belonging to the HR network.

Step 3 — Calculate Sales subnet capacity

Sales uses /26. Host bits = 32 − 26 = 6. Total addresses = 64 and usable hosts = 62. The mask is 255.255.255.192 and the block size is 64. HR has used addresses .0–.127 in the 172.16.0.x range, so the next valid /26 boundary is 172.16.0.128.

Sales occupies 172.16.0.128 through 172.16.0.191. Its network address is 172.16.0.128, broadcast address is 172.16.0.191, and usable host range is 172.16.0.129 through 172.16.0.190. The 64-address block is aligned correctly because 128 is a multiple of 64.

Why VLSM is useful

VLSM avoids assigning the same size to every department when their needs are different. A larger department receives more host addresses, while smaller departments use smaller blocks. Correct allocation saves address space and makes future expansion easier. Each subnet must begin at a boundary compatible with its block size.

QUESTION 3 — CONTINUED: IT, ADMIN AND UNUSED ADDRESS SPACE

Step 4 — Allocate the IT subnet

IT uses /27. Host bits = 32 − 27 = 5. Total addresses = 2^5 = 32, and usable hosts = 32 − 2 = 30. The mask is 255.255.255.224 and the block size is 32. Sales ends at 172.16.0.191, so the next available address is .192, which is a valid multiple of 32.

IT is assigned 172.16.0.192/27. Its network address is 172.16.0.192, its broadcast address is 172.16.0.223, and its usable host range is 172.16.0.193 through 172.16.0.222. The subnet supports 30 hosts.

Step 5 — Allocate the Admin subnet

Admin uses /28. Host bits = 32 − 28 = 4. Total addresses = 2^4 = 16, and usable hosts = 16 − 2 = 14. The subnet mask is 255.255.255.240 and the block size is 16. IT ends at .223, so the next available boundary is .224, which is divisible by 16.

Admin is assigned 172.16.0.224/28. Its network address is 172.16.0.224, broadcast address is 172.16.0.239, and usable host range is 172.16.0.225 through 172.16.0.238. Admin can assign up to 14 ordinary host addresses.

Department	Prefix / mask	Network	Usable hosts	Broadcast	Capacity
HR	/25 255.255.255.128	172.16.0.0	172.16.0.1–.126	172.16.0.127	126
Sales	/26 255.255.255.192	172.16.0.128	172.16.0.129–.190	172.16.0.191	62
IT	/27 255.255.255.224	172.16.0.192	172.16.0.193–.222	172.16.0.223	30
Admin	/28 255.255.255.240	172.16.0.224	172.16.0.225–.238	172.16.0.239	14

Step 6 — Verify the allocation

The HR subnet ends at .127 and Sales begins at .128. Sales ends at .191 and IT begins at .192. IT ends at .223 and Admin begins at .224. These consecutive boundaries demonstrate that the subnets do not overlap. Their block sizes are 128, 64, 32, and 16 addresses, respectively, matching the requested prefix lengths.

Step 7 — Calculate the remaining address range

The four subnets use 128 + 64 + 32 + 16 = 240 addresses. The /16 block contains 65,536 addresses, so the number left unused is 65,536 − 240 = 65,296. The assigned subnets end at 172.16.0.239; therefore the next unallocated address is 172.16.0.240. The remaining range extends from 172.16.0.240 through 172.16.255.255, assuming no other allocations exist.

Conclusion

The four departmental networks are correctly sized and non-overlapping. HR supports 126 hosts, Sales supports 62, IT supports 30, and Admin supports 14. VLSM uses 240 addresses for the stated requirements and leaves 65,296 addresses available for future networks.

QUESTION 4 — SUBNET 172.16.10.0/24 INTO /27

Problem statement and given information

A company has network **172.16.10.0/24** and changes the subnet mask to **/27**. Determine the number of subnets created, the usable host capacity of each subnet, the subnet containing 172.16.10.78, and whether 172.16.10.95 is a valid host address in that subnet.

Step 1 — Calculate the number of borrowed bits

The original prefix is /24 and the new prefix is /27. The difference is 27 − 24 = 3 bits. These three bits are borrowed from the host portion to create subnet identifiers. The number of possible subnets is 2^3 = 8. This means the original /24 network is divided into eight equal-sized /27 networks.

Step 2 — Find the new subnet mask

The /27 mask contains 27 binary ones followed by five zeros. In dotted-decimal notation it is 255.255.255.224. The last octet 224 is 11100000 in binary, which confirms that three bits of the final octet are used for subnetting and five bits remain for hosts.

Step 3 — Calculate addresses per subnet

The host-bit count is 32 − 27 = 5. Each subnet contains 2^5 = 32 total addresses. Subtracting the network and broadcast addresses gives 32 − 2 = 30 usable host addresses per subnet. Therefore, eight subnets are available and each subnet can accommodate 30 ordinary IPv4 devices.

Step 4 — Calculate the block size and subnet ranges

Block size = 256 − 224 = 32. The final-octet subnet boundaries are 0, 32, 64, 96, 128, 160, 192, and 224. Every subnet spans 32 consecutive addresses. These ranges make it possible to identify the correct network without converting every address to binary.

Subnet	Address range	Network	Broadcast
1	172.16.10.0–.31	172.16.10.0	172.16.10.31
2	172.16.10.32–.63	172.16.10.32	172.16.10.63
3	172.16.10.64–.95	172.16.10.64	172.16.10.95
4	172.16.10.96–.127	172.16.10.96	172.16.10.127
5	172.16.10.128–.159	172.16.10.128	172.16.10.159
6	172.16.10.160–.191	172.16.10.160	172.16.10.191
7	172.16.10.192–.223	172.16.10.192	172.16.10.223
8	172.16.10.224–.255	172.16.10.224	172.16.10.255

QUESTION 4 — CONTINUED: CHECK THE GIVEN IP ADDRESSES

Step 5 — Find the subnet containing 172.16.10.78

The relevant subnet ranges are 64–95 and 96–127. The last octet of the given address is 78, and 78 lies between 64 and 95. Therefore, the address belongs to the subnet 172.16.10.64/27. The network address is 172.16.10.64 and the broadcast address is 172.16.10.95.

The valid host range is found by excluding the network and broadcast addresses. The first usable address is 172.16.10.65, and the last usable address is 172.16.10.94. Since 172.16.10.78 lies between these values, it is a valid host address and can be assigned to a computer, provided that the address is not already in use.

Step 6 — Check whether 172.16.10.95 is a valid host

The address 172.16.10.95 is the final address in the range 172.16.10.64 through 172.16.10.95. By definition, the final address of this conventional IPv4 subnet is the broadcast address. It is used to send traffic to all hosts on the subnet and must not be assigned to an ordinary computer or network interface.

Although .95 falls numerically inside the subnet's range, it is not a valid host address. This is an important distinction: a subnet range contains the network identifier, usable host addresses, and broadcast address. Only the middle addresses can be assigned to ordinary hosts.

Step 7 — Summarize the calculation

Item	Calculation / result
Original network	172.16.10.0/24
New prefix	/27
Borrowed bits	27 − 24 = 3
Number of subnets	$2^3 = 8$
Host bits per subnet	32 − 27 = 5
Total addresses per subnet	2■ = 32
Usable hosts per subnet	32 − 2 = 30
Subnet containing .78	172.16.10.64/27
Network address	172.16.10.64
Valid host range	172.16.10.65–172.16.10.94
Broadcast address	172.16.10.95
Is .78 a valid host?	Yes
Is .95 a valid host?	No; it is the broadcast address

Practical importance

Using /27 subnets can reduce unnecessary address allocation and separate devices into smaller network segments. For example, different departments or floors can use different subnets, which can improve address management and help apply routing and access-control policies. Correctly identifying the broadcast address prevents configuration mistakes and connectivity problems.

Conclusion

The /24 block creates eight /27 subnets, each with 30 usable host addresses. Address 172.16.10.78 is a valid host in 172.16.10.64/27. Address 172.16.10.95 is the broadcast address of that subnet and cannot be assigned to an ordinary host.

QUESTION 5 — IPV6 COMPRESSION, EXPANSION AND /64 PREFIX

Problem statement and given information

Given IPv6 address: **2001:0db8:0000:0000:0000:ff00:0042:8329**. The tasks are to compress the address, expand the compressed form back to its full notation, identify its /64 network prefix, and calculate the number of possible values in the 64-bit interface-identifier space.

Step 1 — Understand IPv6 address structure

IPv6 addresses are 128-bit identifiers written as eight groups of hexadecimal digits. Each group is 16 bits long and is separated from the next group by a colon. These groups are commonly called hextets. An IPv6 address has a much larger address space than IPv4, which uses 32-bit addresses. The address supplied in this problem contains eight hextets, so it is already in full-length notation.

The hexadecimal digits range from 0 to 9 and A to F. Each hextet can contain up to four hexadecimal digits. Leading zeros in a hextet may be omitted without changing its numeric value. For example, 0db8 can be written as db8, and 0042 can be written as 42.

Step 2 — Remove leading zeros

Start with the original address: 2001:0db8:0000:0000:0000:ff00:0042:8329. Remove only the zeros at the beginning of each individual hextet. The address becomes **2001:db8:0:0:0:ff00:42:8329**. The zero-valued hextets are still present at this stage; only leading zeros inside each group have been removed.

This first step is safe because removing leading zeros does not change the value of a hexadecimal group. It is important not to remove zeros from the end of a hextet because that would change the numeric value. For example, 0042 and 42 are equal, but 4200 and 42 are not equal.

Step 3 — Compress the longest consecutive run of zero hextets

IPv6 permits a double colon (::) to replace one consecutive sequence of one or more all-zero hextets. In the supplied address, three zero hextets occur consecutively after 2001:db8. Replace these three zero groups with a single double colon. The compressed address is **2001:db8::ff00:42:8329**.

The double colon must be used no more than once in a single address because using it twice would make it impossible to determine how many omitted zero groups belong at each position. In this example, it represents exactly three zero hextets. Compression changes only the written representation, not the actual 128-bit address.

Step 4 — Confirm the compressed form

The compressed address shows five explicit groups: 2001, db8, ff00, 42, and 8329. IPv6 requires eight hextets in total. Therefore, the double colon represents $8 - 5 = 3$ missing groups. Restoring three zero groups gives the original eight-group address. This confirms that the compression is correct.

QUESTION 5 — CONTINUED: EXPANSION, PREFIX AND CAPACITY

Step 5 — Expand the compressed address

The compressed form is 2001:db8::ff00:42:8329. There are five visible hextets, so the double colon stands for three all-zero hextets. Add those three groups in the position indicated by the double colon and restore four digits per hextet where necessary. The full form is **2001:0db8:0000:0000:0000:ff00:0042:8329**.

The expanded address matches the address given in the question. This verification is useful when troubleshooting IPv6 configurations because it confirms that no group has been added, removed, or moved incorrectly.

Step 6 — Identify the /64 network prefix

A /64 prefix means that the first 64 bits identify the network prefix. Since each hextet contains 16 bits, the first four hextets make up 4 × 16 = 64 bits. The first four hextets are 2001:0db8:0000:0000. Therefore, the /64 network prefix is **2001:db8:0:0::/64**, which may also be written as 2001:0db8:0000:0000::/64.

The remaining four hextets form the lower 64 bits of this particular address: 0000:ff00:0042:8329. In a common /64 subnet arrangement, the first 64 bits identify the subnet and the remaining 64 bits identify an interface within that subnet. IPv6 addressing architecture and textual representation are described in the Internet standards documentation. Reference: RFC 4291, IP Version 6 Addressing Architecture.

Step 7 — Calculate the number of possible interface identifiers

IPv6 has 128 bits in total. For a /64 prefix, the number of bits left after the network prefix is 128 − 64 = 64. Each bit can have one of two values, 0 or 1. Therefore, the number of possible 64-bit values is 2^{64} = **18,446,744,073,709,551,616**.

This number describes the size of the 64-bit value space. It should not be interpreted as a promise that every value will be assigned to a device. Actual interface addressing depends on network configuration and IPv6 rules. Unlike ordinary IPv4 subnet calculations, IPv6 does not use the IPv4 network-and-broadcast subtraction rule in the same way; IPv6 has no broadcast addresses, and multicast serves related group-delivery purposes.

Final result

Requirement	Answer
Original IPv6 address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed address	2001:db8::ff00:42:8329
Expanded address	2001:0db8:0000:0000:0000:ff00:0042:8329
Network prefix	2001:db8:0:0::/64
Network-prefix bits	64
Interface-identifier bits	128 − 64 = 64
Possible 64-bit values	2■■ = 18,446,744,073,709,551,616

OVERALL CONCLUSION

This assignment demonstrates the practical use of IPv4 subnetting, variable-length subnet masks, host-capacity calculations, network and broadcast identification, and IPv6 address compression. In Question 1, three laboratory subnets are allocated without overlap. In Question 2, two devices are confirmed to belong to the same /26 subnet. In Question 3, four departments receive correctly sized VLSM subnets and the remaining address space is identified. In Question 4, a /24 network is divided into eight /27 subnets and two addresses are classified correctly. In Question 5, the IPv6 address is compressed and expanded, its /64 prefix is identified, and the 64-bit interface-identifier capacity is calculated.

The common method in all five problems is to write the given data, select the correct prefix or mask, calculate the block size and host bits, determine the address boundaries, verify the result, and explain what the result means for a real network. Accurate subnetting supports efficient address use, easier administration, and reliable communication between networked devices.